The Mathematical Economics of Unconventional Monetary Policy

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Abstract

This paper develops a unified mathematical framework for Unconventional Monetary Policy (UMP) by integrating macroeconomics, physics, information theory, actuarial science, and multi-agent strategic systems. We model the economy as a Relativistic-Thermodynamic General Equilibrium (RTGE) constrained by planetary geometry. Central banks process reality via the General Theory of Institutional Signal Simplification (GTISS), utilizing horizontal and vertical rolling windows that inevitably distend under stochastic volatility, bounded by an Institutional Nyquist Frequency. We prove that UMP signals, when fed back into the physical economy, create non-commutative operator discrepancies. Furthermore, we conceptualize the central bank as a macro-insurer utilizing a General Theory of Insurance, where UMP acts as a premium to minimize a global uncertainty functional. We introduce a Quantum-inspired Asset Pricing model incorporating radial accessibility premia, linking financial valuation directly to thermodynamic and relativistic constraints. The framework encompasses welfare economics, geopolitical warfare, psychology, and artificial intelligence, providing a rigorous foundation for policy optimization.

The paper ends with “The End”

1 Introduction

Unconventional Monetary Policy (UMP)–encompassing quantitative easing, forward guidance, and negative interest rates–operates at the boundary of macroeconomic theory, financial physics, and actuarial risk management. Traditional models abstract from physical constraints, information processing limits, and systemic risk pooling. This paper synthesizes insights from planetary physics, thermodynamics, relativity, institutional signal processing, actuarial science, and quantum finance to construct a robust general equilibrium framework. We demonstrate that the economically actionable domain is a bounded planetary shell, and that central bank communication inherently acts as a lossy signal compressor, generating structural vulnerabilities. Concurrently, UMP functions as a systemic insurance mechanism, priced according to marginal risk contributions.

2 The Macro-Physical State Space (RTGE)

Consider a spherical planet of radius R. At every surface point $x \in S^2(R)$, there exist two radial directions: the inward geological normal $-n(x)$ and the outward astronomical normal $+n(x)$. The purely geometric radial domain is $-R \leq z < \infty$, but the economically accessible domain is strictly bounded.

Definition 1 (Planetary Economic Shell)**.** *The economically accessible domain at time t is defined as $-D_G^*(t) \leq z \leq D_A^*(t)$, where D_G^* is the maximum geological depth and D_A^* is the maximum astronomical distance.*

Theorem 1 (Relativistic-Thermodynamic Radial Asymmetry)**.** *Under finite technological resources, increasing geothermal gradients, and the relativistic speed limit $v < c$, the economically accessible radial opportunity set is finite and strictly interior to the physical boundaries: $D_G^*(t) < R$ and $D_A^*(t) < \infty$.*

Proof. Geological extraction costs C_G increase with depth d, temperature $T(d)$, and pressure $P(d)$. Since $T'(d) > 0$ and $P'(d) > 0$, extraction becomes prohibitive at $D < R$, hence $D_G^* \leq D < R$. Outward, the present value of a project at distance D is bounded by $PV(D) \leq X(D)e^{-2\rho D/c}$. As $D \to \infty$, $PV(D) \to 0$. Thus, $D_A^*(t) < \infty$. □

3 Quantum Asset Pricing and UMP

Traditional asset pricing ignores physical origin. We classify assets into in-planet (I), surface (S), and off-planet (E) classes. Let $|R_i\rangle$ represent radial origin and $|F_i\rangle$ represent conventional financial characteristics. The asset state is:

$$|\Psi_i\rangle = |R_i\rangle \otimes |F_i\rangle$$

Under UMP, the central bank alters the stochastic discount factor (SDF) $M_{t,t+\tau}$. We extend the Capital Asset Pricing Model (CAPM) to include an accessibility premium λ_A:

$$E(R_i) - r_f = \beta_i \lambda_M + \gamma_i \lambda_A$$

where γ_i measures exposure to physical accessibility risk (thermodynamic or relativistic). The covariance matrix Σ extends naturally across radial sectors, requiring portfolio diversification across I, S, and E.

4 Institutional Signal Simplification (GTISS)

Central banks do not observe the true macro state $X_t \in \mathbb{R}^n$. They publish a compressed signal $S_t = C(X_t)$ using two operators: horizontal (time) and vertical (peer/hierarchy).

Definition 2 (Operators). *The horizontal operator is*

$$H_w[X]_{i,t} = \frac{1}{w} \sum_{s=0}^{w-1} X_{i,t-s}$$

.

The vertical operator is

$$V_k[X]_{i,t} = X_{i,t} - \bar{X}_{N_k(i,t)}(t)$$

.

Definition 3 (Institutional Nyquist Frequency). *Signal processing dictates that no hierarchy can accurately detect changes occurring above a critical frequency:*

$$f_N = \frac{1}{2(H_h + H_v)}$$

where H_h and H_v are horizontal and vertical memory horizons. Changes with $f > f_N$ are aliased, causing the institution to misinterpret reality.

Proposition 1 (Non-Commutativity under Stochastic Volatility). *Horizontal and vertical operators generally do not commute: $C_{i,t} = (H_w V_k - V_k H_w)[X]_{i,t} \neq 0$. If the peer set $N_k(i,t)$ is static, $C_{i,t} = 0$. Under peer turnover rate γ,*

$$E[C_{i,t}^2] = \Theta(\frac{\gamma w \sigma_X^2}{k})$$

.

Proof. With a static peer set, linearity allows the exchange of summation over time s and peers $N_k(i)$. With turnover, a fraction ρ_s of peers differ at lag s, leaving a residual variance of approximately $2\rho_s\sigma_X^2/k$. Summing over independent lags yields the Θ rate. □

When published signals feed back into the economy

$$dX_t = \beta S_t dt + \sqrt{V_t} dW_t$$

two independent failure modes emerge: resonance instability from window mismatch ($\beta < 1 + w\gamma$) and amplitude instability from volatility-of-volatility ($2\kappa > \xi^2$).

5 A General Theory of Insurance and UMP

The central bank functions as a macro-insurer, transforming uncertain systemic losses into predictable policy operations. We define a Unified Risk Functional encompassing multiple uncertainty layers.

Definition 4 (Risk Structures)**.** *For a population of N agents, individual uncertainty is defined as:*

$$R_i = \alpha A_i + \beta U_i + \gamma S_i + \delta M_i$$

where

$$A_i = E[\mathit{Var}(L_i|\theta_i)]$$

is physical risk,

$$U_i = \mathit{Var}(E[L_i|\theta_i])$$

is actuarial risk,

$$S_i = \mathit{Cov}(L_i, L_T)$$

is systemic risk, and

$$M_i = \mathit{Var}(E[L_i|F^{(k)}])$$

is model risk.

Aggregate uncertainty becomes $\mathcal{R} = \lambda_1 \sum_{i=1}^{N} R_i + \lambda_2 R_T$. The central bank minimizes this global functional.

Theorem 2 (General Premium Principle)**.** *The optimal UMP intervention (premium) satisfies:*

$$P_i = \frac{\delta \mathcal{R}}{\delta L_i}$$

Proof. Suppose total premium revenue (seigniorage or taxation capacity) is constrained by solvency requirements. The insurer minimizes

$$J = \mathcal{R} + \lambda(\sum_i P_i - K)$$

. First-order optimality conditions imply

$$\frac{\partial J}{\partial P_i} = 0$$

. Therefore, each premium equals the marginal increase in total uncertainty generated by the corresponding insured agent. □

This insurance mechanism is also a cooperative game. Using the Shapley allocation

$$P_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(N - |S| - 1)!}{N!} [v(S \cup \{i\}) - v(S)]$$

the central bank fairly allocates uncertainty costs. Bayesian updates reduce actuarial risk U_i as policy data D generates posterior $\pi(\theta_i|D)$.

6 Interdisciplinary Implications

6.1 Physics, Chemistry, and Biology

UMP is anchored in physical reality. Thermodynamics limits the extraction of real resources backing the fiat economy, as entropy production caps conversion efficiency $\eta = W_u/E$. Biologically, human cognitive filters act as vertical rolling windows, creating structural blindness to systemic shocks (e.g., the 2020 liquidity crisis).

6.2 Psychiatry and Philosophy

Long-term UMP isolation creates psychiatric stress. Effective labor becomes

$$L_{\text{eff}} = L\Psi(\tau, \sigma, I)$$

where communication delay τ, stress σ, and isolation I reduce cognitive performance. Philosophically, UMP transforms systemic risk into permanent unknowns via vertical information ceilings, conforming to the ontological hierarchy: existence $\supseteq$ observability $\supseteq$ technical accessibility $\supseteq$ economic accessibility.

6.3 Political Science, Geopolitics, and Diplomacy

UMP acts as financial warfare. By altering the discount rate ρ, a hegemon can strategically deny capital to adversaries. Sovereign credit ratings function as vertical operators, structurally blind to common-factor geopolitical shocks. Diplomatic communication networks operate under bandwidth constraints B, forcing signal simplification ratio

$$\rho = \frac{E(S_t)}{E(X_t)} \leq 1$$

.

6.4 Warfare Economics and Welfare Economics

During conflict, correlated losses collapse diversification benefits, shifting systemic risk premia. Warfare changes the shadow price of both inward (minerals) and outward (satellites) capital. From a welfare perspective, because published UMP signals are garblings, any policy conditioned on S_t rather than X_t yields a welfare loss by Blackwell's theorem. The central bank optimizes intergenerational welfare

$$W = \sum_{g=0}^{\infty} \beta^g U_g$$

.

6.5 Statistics, Computer Science, and AI/ML

Central banks increasingly utilize reinforcement learning to optimize policy under uncertainty. The state vector

$$s_t = (X_t, V_t, \mathcal{I}_t)$$

is mapped to an action a_t to maximize

$$V(s) = \max_a \{r(s, a) + \beta E[V(s')]\}$$

. Data science pipelines construct features from raw signals, acting as institutional compression operators.

Table 1: Interdisciplinary Synthesis of UMP Mechanisms

Field	Mechanism	Mathematical Object	Economic Consequence
Physics	RTGE	$-D_G^* \leq z \leq D_A^*$	Bounded economic opportunity
Finance	Quantum Pricing	$\lvert\Psi_i\rangle = \lvert R_i\rangle \otimes \lvert F_i\rangle$	Accessibility premium λ_A
Info Theory	GTISS	$f_N = 1/(2(H_h + H_v))$	Aliased macro signals
Actuarial	General Insurance	$P_i = \delta\mathcal{R}/\delta L_i$	Marginal risk pricing
CS/AI	RL Bellman	$V(s) = \max_a\{r + \beta E[V]\}$	Algorithmic policy

7 Algorithmic Representation

The following pseudo-code outlines the central bank's policy optimization algorithm integrating RTGE, GTISS, and General Insurance frameworks.

```
INPUT: State vector X_t, Peer set N_k, Window lengths w, H_v, Loss vector L
OUTPUT: Policy action a_t (UMP parameters), Premium P_i

1. Evaluate physical constraints: compute D*_G and D*_A
2. Compute ISS signal: S_t = V_k(H_w(X_t))
3. Check Institutional Nyquist Frequency: if signal_freq > f_N, flag aliasing
4. Calculate risk components: A_i, U_i, S_i, M_i
5. Construct global uncertainty functional: R = lambda_1 * sum(R_i) + lambda_2 * R_T
6. Compute UMP premium: P_i = delta_R / delta_L_i
7. Update volatility estimate V_t via GARCH-diffusion filter
8. Evaluate welfare function: W = U(C_t) - Psi(E_t) - Crisk
9. Optimize action: a_t = argmax_{a} { r(s_t, a) + beta * E[V(s') | s_t, a] }
10. Execute UMP transmission: broadcast S_t = a_t to radial economy
```

8 Vector Graphics

The following PGF/TikZ diagram illustrates the feedback loop between the central bank's signal simplification, the insurance risk functional, and the planetary economic shell.

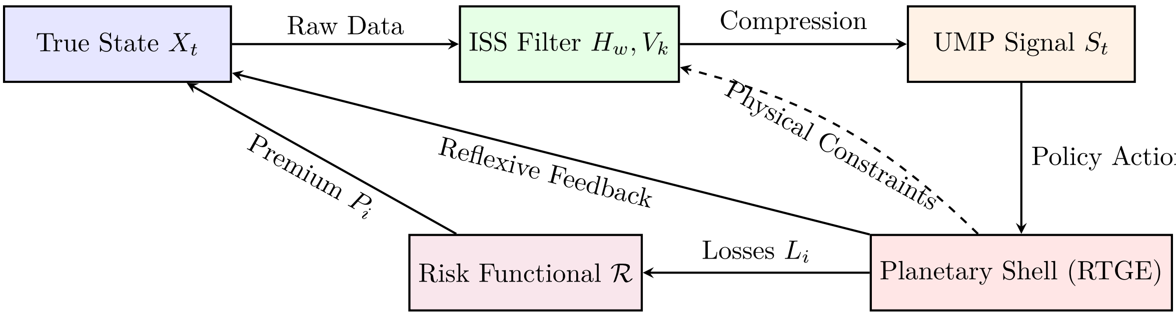

Figure 1: Architecture of UMP Transmission.

9 Conclusion

Unconventional Monetary Policy cannot be understood purely through traditional monetary economics. By embedding UMP within a Relativistic-Thermodynamic General Equilibrium, modeling central bank communication as Institutional Signal Simplification, and pricing interventions via a General Theory of Insurance, we reveal the structural limits of policy efficacy. The non-commutativity of temporal and peer-group averaging, combined with physical constraints and systemic risk pooling, guarantees that UMP operates under irreducible information loss and structural risk ceilings. Future research must integrate these physical, information-theoretic, and actuarial bounds into the next generation of macroeconomic models.

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Glossary

Accessibility Premium (λ_A)
The additional expected return required by investors to compensate for physical accessibility risk associated with an asset's radial origin.

Actuarial Risk (U_i)
The variance of the expected loss conditioned on unknown parameters, reflecting uncertainty in loss distributions.

Commutator ($C_{i,t}$)
The mathematical discrepancy between applying horizontal and vertical averaging operators in different orders; driven by peer turnover.

Institutional Nyquist Frequency (f_N)
The critical frequency threshold beyond which institutions misinterpret reality due to structural lag.

Planetary Economic Shell
The bounded radial domain $[-D_G^*, D_A^*]$ representing the economically accessible physical space around a planet.

Quantum State ($|\Psi_i\rangle$)
A mathematical representation of an asset combining its radial origin and financial characteristics.

Unified Risk Functional ($\mathcal{R}$)
A global measure of uncertainty encompassing physical, actuarial, systemic, and model risks.

Vertical Operator (V_k)
A cross-sectional averaging or demeaning filter applied against a reference group of k peers.

The End